Chapter 10

Upgrading and refurbishment systems

Even the newest and most advanced of control systems eventually becomes dated. Up to about the middle of the twentieth century, C&I systems were mainly pneumatic, the pace of development was slow and it was generally possible to keep a system in good functional order, however old it may have been, simply by repairing the equipment or, if the worst came to the worst, buying replacement components from the original manufacturer. (In those heady days manufacturers could be relied on to remain in business for more than a few years, and to be able and willing to provide replacement components at reasonable cost for as long as the plant itself remained in operation!)

This happy situation persisted for a while after the first (analogue) electronic systems arrived, but the advent of computer-based systems eventually brought about a sea-change in the maintenance of equipment and systems.

For the hapless end-user, it often seemed that as soon as a new system was installed it had become obsolete. To make matters worse, the replacement was not only better, faster and cheaper but it was so widely used that it quickly began to be difficult to find maintenance staff who were able to understand and work with the 'old fashioned' equipment that had so recently been the owner's pride and joy. Worse still, as time went by the replacement parts became steadily more expensive and difficult to obtain.

The ultimate disaster would be when the manufacturer of the system was taken over or went out of business entirely. In the takeover situation the new proprietor would sometimes try and persuade the end user to replace the old system with one from his own stable. If the pressures were resisted, maintenance of the old system could become very expensive indeed!

It's not altogether a desirable situation, but it's what we have, so in this chapter we shall look at the problem and provide some advice on how to live with it. We start with a quick look at the reasons behind the situation, the intention being to show that, rather than being due to manufacturers trying to bolster their sales revenues, there are legitimate reasons for the constant evolution of systems and technologies.

10.1 The reasons behind the changes

The development of a computer-based system is a very complex, time-consuming and expensive business. The need for a new system is usually brought about by a competitor offering something better and cheaper (or at least apparently more cost-effective), and when that happens sales of the present generation begin to collapse. Clearly, if this situation is allowed to persist it will ultimately lead to the demise of the company, so a project has to be launched to produce the next-generation system.

At this point the systems manufacturer must look at the latest offerings from the manufacturers of the microprocessors—the components that are at the heart of all modern systems. For such companies, industrial process-control systems are not among the most valuable applications, because sales into this market are small in volume yet face the most stringent requirements for performance, reliability and safety.

The manufacturer of a domestic washing machine will buy very many more devices and will be far less demanding with respect to the performance and reliability requirements.

Reliability and consequential loss are major issues in the process-control market. A washing machine that lasts for five years seems to be almost the exception these days, whereas power stations have to operate over a lifetime of 20 years or more. A failed device on a washing machine may in the long term result in compensation claims, but these are likely to be small in comparison with those resulting from an incident in a power station. When the plant is a nuclear power station, the costs can become enormous, and the implications of failure are daunting. (After the incident at Unit 2 of the Three Mile Island nuclear power plant in the USA on 28 March 1979, one minicomputer manufacturer placed a complete and worldwide embargo on the use of their products on any nuclear plant. This presented something of a problem to the many C&I systems vendors who were already completing the manufacture of systems based on these products, for installation on nuclear power stations around the world.)

Once the attention of the device manufacturer has been captured, the C&I system manufacturer has to persuade him to divulge the development

plans for his product range. The reason for this is simple: the time-scale involved in developing and launching a new computer-based system may be two or three years. This time-scale is comparable with the entire life expectancy of a single generation of microprocessors. In other words, if the C&I systems manufacturer launches a development programme based on a variety of microprocessor that is available today, the device will be obsolete by the time the system is launched.

The C&I manufacturer therefore has to start designing his system and building prototypes using early development samples of microprocessors and so forth. At this stage his designs must be based on a great deal of predicted data, some of which may prove to be overoptimistic.

At the end of all this, the new system is launched and after early trials it becomes widely used (or so the supplier hopes). Because of this success, the manufacturer of a rival system then begins to lose his own market share, so he too starts looking at a new development. This is eventually launched. A complex race, an expensive game of 'technical leap-frog', is then on.

Even without this competition, each new generation of microprocessor offers advances in terms of power and performance, and offers solutions to problems that usage of the previous generation has exposed. Worse still, the manufacturers of the electronic components cannot support endless varieties of devices, so the day eventually arrives when the older generation of components is withdrawn. This adds further complexity to the life of the C&I manufacturer who has a responsibility to keep supplying his customers with replacement parts over several decades.

10.2 Living with change

If the above sounds like an excuse, it is, but it is a very valid excuse, because the pressures are real, and they are inexorable. The only realistic approach therefore, is to accept that change is inevitable and to formulate a plan to deal with it. All the same, when a system is purchased every possible precaution should be taken to ensure that it provides satisfactory service over a long period. Some precautions are listed below:

10.2.1 Defining the requirements for long-term support

As might be expected, much depends on the actions that are taken at the outset, even before the system has been ordered. At the definition stage of a DCS contract an important target is to wring from the system vendor as many written assurances of long-term support as possible, and to make as

sure as is possible that the vendor is reliable and that he will be able to stand by his promises. It is a hard goal to meet, but the time to press for such promises is when an order for several hundreds of thousands of pounds, sometimes millions, is in the offing.

Unfortunately, the commercial clauses in large contracts are usually framed by people who know little or nothing about the evolution of DCS or C&I technology and the effects on plant maintenance. In this situation a vendor can respond with weasel words that are worth very little in reality.

Having obtained the promises, it is vital that they are examined as carefully as possible by professionals who really understand the relevant technology. The time and cost of this evaluation will be repaid many times, over the operational life of the plant.

10.2.2 Keeping track of costs

Once the equipment has been acquired, the users must focus on how best to use it, and they must be prepared to ignore all information on new systems (which are always better than the one that has just been bought!). The fact is that it has been bought, and now the users must gain the maximum advantage from the investment.

An important procedure is to keep careful records of all the markers along the road that stretches from initial acquisition to ultimate replacement:

- Log each failure that occurs, recording the time and date of the event. This is useful information for the user and the vendor, since it supports or disproves reliability claims.
- Keep a note of the time that elapses between placing an order and receiving the goods. This will help reinforce your arguments that delays in shipping spares are increasing, exposing the plant to increased levels of risk, causing reduced output, or incurring increased operational costs.
- Monitor and record how long it takes to replace the failed item of equipment with a spare from stock.
- Record the cost of spare parts (printed-circuit cards, power supplies etc.).
- Note the delivery time for spares.
- When recording details of a failure, make a note of any plant outage or reduction in output that can be directly attributed to the failure.
- It is also useful to try and relate the failure to any event that occurred at or just before the time at which the failure occurred, such as a severe thunderstorm, or the failure of any other electrical plant or machines. This can assist in post-mortem analyses which may point to a design

defect such as inadequate screening or earthing, or poor design of power supplies.

Table 10.1 shows a simple format that can be used for recording this type of information.

10.3 Making the decision to change

The day will eventually dawn when it is wise to examine the arguments for and against upgrading the DCS, and to then decide on the best course of action to take. Reaching a decision will be eased if careful records have been maintained as described above, but even then pressures will be applied to upgrade the system.

10.3.1 The impact of change on the operators

System refurbishment can produce major changes to the operator interfaces. Such changes are greatest when a system based on the use of a hard-wired desk is replaced with a so-called 'soft desk'.

In the former, every actuator is controlled via its own discrete hand/auto station (alternatively called an auto/manual station). These enabled the operator to completely disconnect the electronic controller from the actuator and to manually position the final element via a discrete control circuit. Such a system provided a high degree of redundancy since, with the exception of the wires linking the control desk to the actuator, the circuit for providing manual control was totally separate from the electronics of the automatic controller.

Hard desks also provided discrete indicators (often moving-coil meters) showing the important plant parameters. Sometimes these indicators were driven by transmitters which were completely separate from those feeding the electronic controller, again providing a high degree of redundancy.

With the advent of high-reliability systems arguments began to be raised that the cost of providing such levels of redundancy was no longer justified. But it was only with the advent of sophisticated fault-detection systems, coupled with highly reliable control computers (often embodying two-out-of-three fault-detection systems) that the more conservative engineers began to accept that it was safe to entrust the control of large-scale hazardous processes to a soft-desk concept.

With boiler control systems, even if such a decision is reached, care should be taken to ensure that essential information is available to the operator in the event of a serious failure of the control system or its power-

supply system. For example, there is a need to provide density compensation for certain measurements if a DCS failure occurs. (An example is the drum-level measurement, where the requirements for compensation in the presence of a DCS failure were pointed out in Section 6.3.)

10.4 A refurbishment case study

Changing to a 'soft' desk results in the operator being presented with a system of displays and controls which is completely different from that with which he or she was familiar when using the hard-wired predecessor.

It may be useful to briefly examine a particular example, the control desk and console of a large coal-fired boiler. The original control-room facilities had been supplied in the days of analogue electronic systems (*c.* 1975) and had vertical panels providing full indication and control facilities of all eight mills, plus a multiplexing system that enabled the operator to gain access from the desk to any two of the mills at a given time. At the time of the original design, this presented some difficulties for ensuring that gaining control of a pair of mills did not introduce any disturbance into the plant.

This system was replaced by a distributed system in 1989 (Figure 10.1) but this was, in turn, upgraded with more up-to-date systems in 1998. Figure 10.2 shows the new 'soft-desk' control-room facilities after the second refurbishment.

The use of screen-based control-room facilities enables a wealth of information to be provided for the operator in a form that is easy to assimilate and use. A 'soft desk' is also inherently more flexible, allowing additional control and display facilities to be easily added as requirements evolve or new plant is added.

10.4.1 The involvement of plant staff

Experience has shown that the success of 'soft desk' upgrades can be correlated to the level of ownership taken by the unit operators. The current strategy of many of the DCS & SCADA implementers is to involve the operators in the process of designing the mimic displays, so that they are happy to use the new screens and thus take ownership in the success of the scheme.

In any case, dramatic changes such as these necessitate a programme of retraining for all plant personnel. The operators must learn how to use the new display and control facilities and the maintenance staff must be

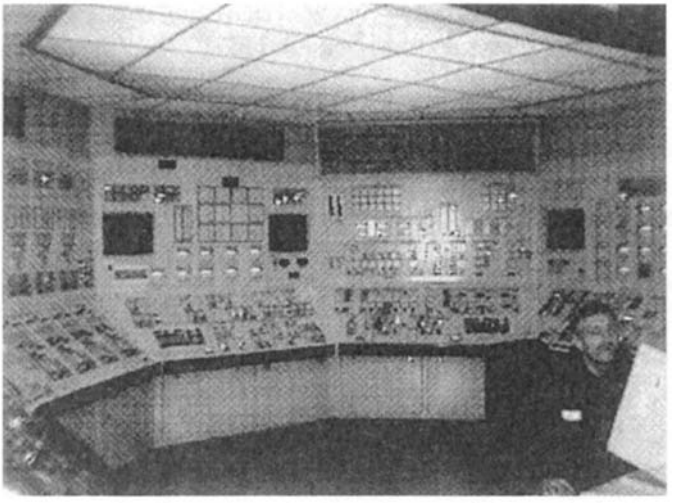


Figure 10.1 *The 'hard-wired' control console and panel of a large coal-fired power-plant before refurbishment*
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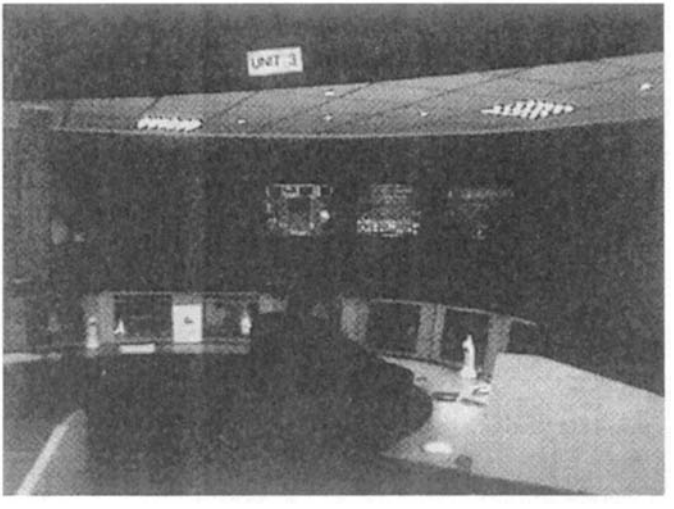


Figure 10.2 *A 'soft desk' replacement for the 'hard-wired' operator facilities after refurbishment*
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trained in the maintenance of the new systems. It is of course essential that the costs of such training are considered in the cost/benefit analysis of a refurbishment project.

10.5 Why refurbish?

Plant managers will be constantly besieged by DCS sales personnel who will be only too happy to provide plenty of arguments to justify replacement of the existing systems with their own. To counterbalance this weight of evidence, here are a couple of examples of reasons why a system should not be upgraded.

10.5.1 'The performance of the present system has degraded'

An operational control system that has at one time met its performance requirements is unlikely to start producing inferior results merely because of deterioration of the electronic components. A component failure may occasionally result in drift, or it may cause the system to operate wrongly, but the solution in such cases is to replace the component or the card in which it is fitted.

Deterioration of a DCS's ability to control the plant is far more likely to be the result of poor performance of the interface transducers (the transmitters and actuators), rather than any factor in the system itself. For example, inadequate maintenance of a valve actuator will result in hysteresis or overshoot. Changing the electronic system that controls the valve will do nothing to improve on control performance.

Equally likely is the possibility that the controlled element itself (a valve or damper) requires maintenance. In several cases, the application of grease to a shaft or bearing has worked wonders on the performance of the loop!

The point is this: if the performance of an actuator or a final element has deteriorated because of lack of maintenance, it is unlikely that a major investment in a new DCS will result in any dramatic long-term improvement. It was the attitudes of the management or staff that resulted in the lack of maintenance and these will not necessarily change when the new system arrives. A short-term improvement may well be obtained at the outset (because the manufacturer will have been forced to fine-tune the performance of all actuators and final elements in order to meet performance guarantees, and because any new system engenders some enthusiasm among maintenance staff, if not the plant operators), but once

the situation has settled down the lack of maintenance will once again result in an eventual degradation in performance.

10.5.2 New systems are faster, more powerful, better

Users will always be aware of a system that seems to work better in some way than the one installed on their own plant. That's the nature of the game.

But the system installed on any installation has been adjusted to work effectively with the idiosyncrasies of the plant. It will have taken somebody a long time to reach that stage. If the system is changed that learning curve will have to be renegotiated when the system is replaced. This underlines the importance of documentation, as described in the following section.

10.6 Documenting the present system configuration

It is almost inevitable that the design of a control system will alter between the initial conceptual stage, through handover and during prolonged use afterwards. The initial system configuration will be carefully defined before commissioning begins. Although the subsequent changes may be recorded, all too often the reasons for introducing them may not be logged. Although the changes may be the result of errors, it is equally likely that they are required because of some misunderstanding, during the design phase, of the plant's characteristics or functions.

The changes are implemented in order to correct such errors or misunderstandings, so that the system works better. But if the reasons for the changes are forgotten, there is every likelihood that when the time comes to refurbish or replace the systems the original misunderstandings will be repeated.

When embarking on a replacement or refurbishment project, the task will be immeasurably assisted by the availability of comprehensive and accurate documentation of the systems as finally configured, together with detailed information on the reasons behind all changes to the original designs.

10.7 Summary

We have now reached the end of this overview of a wide and complex subject. I hope that I have been able to throw some light on the technology

and that the explanations may have lifted some of the veils of mystery that sometimes seem to obscure it. The fact remains that it is a complex matter and it is unwise to entrust the safety of a plant to people who do not understand either the control aspects or the plant operations.

It has been my privilege to have worked with power plant throughout my career and I hope that some of what I have put down here will be useful, and that it may encourage others to take up a very interesting and important subject.

Table 10.1 A form for recording failures and their possible causes, and related factors

Log No.	Failed item		Date of failure	Time of failure	Nature of failure (off calibration, total loss, etc.)	Effects of failure (loss of generation, increased manning levels, cost of additional fuel, resources etc.)	Failure of other plant or machines immediately before or at the same time	Other factors (electrical storms, exceptional heat or humidity etc.)	Replacement			Signatures
	Part No.	Serial No.							Date ordered	Order No.	Date received	
A	A	A	A	A	A	B	C	D	E	F	G	
1	B1288	384723	05 July 98	14:24	Card ouput lost completely	Unit load reduced to 79 MW for 10 hrs	Equipment-room HVAC plant failed 8 hours earlier	Hot day 32°C	06 Jul 98	SP/09/65660	12/8/98	A, B: J Harris C-E: Schmidt G: D Bergoff